Revision Comparison

A grammar edit beside a multiline equation

LANGUAGE-ONLY / SYNTHETIC EXAMPLE

Task. Correct subject–verb agreement. Preserve the setup, dimensions, regularization condition, equation, and concluding statement. This is synthetic material, not an experimental or theorem-validity report.

C1 / Context / editor summary

The passage introduces a regularized least-squares objective. The parameter dimensions and positive regularization condition remain part of the setup shown in both versions below.

C1 / BEFORE — **Anchor: "We minimizes"**

Let $X \in \mathbb{R}^{n\times d}$, $y \in \mathbb{R}^n$, $n \geq 1$, and $\lambda > 0$. The parameter is $\theta \in \mathbb{R}^d$.
We minimizes the following regularized objective:

$$F(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(\sum_{j=1}^{d} X_{ij}\theta_j - y_i\right)^2 + \frac{\lambda}{2}\|\theta\|_2^2 \qquad \text{(B1)}$$
$$= \frac{1}{2n}\|X\theta - y\|_2^2 + \frac{\lambda}{2}\sum_{j=1}^{d}\theta_j^2.$$

The two lines of Equation (B1) express the same objective.

C1 / AFTER — **Anchor: "We minimize"**

Let $X \in \mathbb{R}^{n\times d}$, $y \in \mathbb{R}^n$, $n \geq 1$, and $\lambda > 0$. The parameter is $\theta \in \mathbb{R}^d$.
We minimize the following regularized objective:

$$F(\theta) = \frac{1}{2n}\sum_{i=1}^{n}\left(\sum_{j=1}^{d} X_{ij}\theta_j - y_i\right)^2 + \frac{\lambda}{2}\|\theta\|_2^2 \qquad \text{(A1)}$$
$$= \frac{1}{2n}\|X\theta - y\|_2^2 + \frac{\lambda}{2}\sum_{j=1}^{d}\theta_j^2.$$

The two lines of Equation (A1) express the same objective.

C1 / Reason / editor-proposed

Correct the verb agreement with "We." The subject, action, and mathematical content are preserved. Purple identifies the checked word replacement.

Meaning and review. Exposition-only. The verb ending is the only source change; no proof or experiment has been audited. Formulas remain full-size, outside inline color macros.

Versions and coverage. `before.tex` and `after.tex`; hashes in `identity.json`. Entire fragments compared; C1 is the only text edit. **Presentation mapping.** Source `eq:objective` appears as `before:eq:objective` (B1) and `after:eq:objective` (A1), with separate counters and hyperlink destinations. Both source files remain unchanged. With annotations off, both passages and equations remain visible.